Reconciling the Firm Size and Innovation Puzzle

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Abstract

Since Schumpeter, there has been a lively debate regarding the optimal firm size for innovation. Empirical results have settled into a puzzle: R&D spending increasing with scale, while R&D productivity decreases with scale. Thus large firms appear irrational. We propose and test two alternative resolutions of the puzzle: 1) that it arises from measurement problems, and 2) that firm size endogenously drives R&D strategy, and that the returns to R&D strategies depend on scale. To test both propositions we use recently available NSF BRDIS survey data of firms R&D practices (strategies) as well as a broader measure of R&D productivity. Using the broader measure, we find that both R&D spending and R&D productivity increase with scale—thus offering one resolution to the puzzle. We further find that while large firms and small firms differ in the types of R&D they conduct, there is no type whose returns decrease in scale—there are merely types for which the small firm penalty is less severe. Thus Schumpeter appears to be correct--large firms are the major engine of growth, they both spend more in aggregate than small firms, and are more productive with that spending.

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Reconciling the Firm Size and Innovation Puzzle

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Since Schumpeter, there has been a lively debate regarding the optimal firm size for innovation. Empirical results have settled into a puzzle: R&D spending increasing with scale, while R&D productivity decreases with scale. Thus large firms appear irrational. We propose and test two alternative resolutions of the puzzle: 1) that it arises from measurement problems, and 2) that firm size endogenously drives R&D strategy, and that the returns to R&D strategies depend on scale. To test both propositions we use recently available NSF BRDIS survey data of firms R&D practices (strategies) as well as a broader measure of R&D productivity. Using the broader measure, we find that both R&D spending and R&D productivity increase with scale—thus offering one resolution to the puzzle. We further find that while large firms and small firms differ in the types of R&D they conduct, there is no type whose returns decrease in scale—there are merely types for which the small firm penalty is less severe. Thus Schumpeter appears to be correct--large firms are the major engine of growth, they both spend more in aggregate than small firms, and are more productive with that spending.

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I. Introduction

A common innovation prescription is that firms should behave more entrepreneurially, suggesting that the optimal size for innovation is small firms. So it is interesting that the bulk of R&D in the US is conducted by large established firms-- 3.9% of firms account for 77.3% of R&D (NSF 2003). It is also interesting that this would come as no surprise to Joseph Schumpeter, the economist most often associated with entrepreneurship. Despite all the attention to his "creative destruction", Schumpeter asserted that large firms are the major engine of economic growth (1942). His assertion rested on two arguments. The first argument is that large firms enjoy economies of scale. At the simplest level, this occurs because larger scale amortizes the fixed cost per innovation over more units of output. The second argument is that large firms tend to have greater market power, which means they enjoy higher price premia. Accordingly, both the lower cost and the higher price premia suggest the returns to innovation should be greater for large firms.

Since Schumpeter, there has been a lively debate over which firm size[1] is more amenable to innovation. Indeed, the debate has become one of the most examined questions in innovation economics. The debate is important to economics because innovation is the major driver of economic growth (Solow 1957, Romer 1990). Accordingly, if large firms are more innovative, then perhaps anti-trust policy should be more lenient toward large firms—balancing the innovative benefits of scale against the anti-competitive effects on price. Conversely, if small firms are more innovative, then anti-trust policy has yet another reason to limit industry consolidation.

The debate is also important to firm strategy, because it raises the question of whether there is an optimal firm size for technology firms. If so, once firms grow beyond that size, they may need to decompose into smaller units. Alternatively, if there is an innovative advantage to large scale, then firms may want to aggressively pursue consolidation. Thus it is not surprising that the debate rages in the management literature as well.

Management arguments in favor of large firm innovation (beyond those advanced by

[1] While it seems awkward to convert the continuous size variable into two discrete classes, this follows convention in the theoretical literature, and has an empirical incarnation (the SBA threshold of 500 firms, as used by Acs and Audretsch 1988 and others)

Schumpeter), point to the fact that large firms are more effective conducting R&D. This occurs for a number of reasons. First, there may be minimum efficient scales for some R&D projects, e.g., there is only one super-collider. Second, R&D projects are known to be stochastic. Stevens and Burley (1997) report that on average for Industrial Research Institute (IRI) member firms, it takes 125 funded projects to achieve one commercial success. Large firms are better able to absorb this risk because they can pool it over a broader portfolio of projects. Having a broad portfolio confers another advantage: technical diversity—typically more projects implies a broader set of problems and associated expertise (Nelson 1959). This increases the likelihood of having any required expertise in-house. Finally, large scale implies a broader set of product markets. Drawing again upon the stochastic nature of R&D, this increases the likelihood that projects that fail for a given application, might have applications elsewhere in the firm.

Management arguments in favor of small firm innovation typically rely on governance advantages. One such advantage is that small firm compensation systems are better able to align employee behavior with company goals. First, because there are fewer employees it is easier to observe employee behavior directly. Accordingly, it is easier to design compensation systems based on observed behavior. Second, often there is only one group or department in the company. This means company-level outcomes are a reasonable measure of performance for that group. These compensation advantages translate into both better ability to attract high caliber employees and better ability to motivate them. Indeed Zenger (1994) found that Silicon Valley employees leaving for small firms obtained twice the salary premium of those going to large firms. He also found that small companies were more likely to offer equity and other forms of compensation tied to individual and group performance. As a result, small companies were able to attract higher caliber (based on GPA, degrees, awards, publications and patents) employees, and that those employees worked more hours than employees in large companies.

An additional organizational advantage small companies enjoy beyond the ability to attract higher caliber employees and induce them to work harder, is better communication. Having fewer employees implies decision makers are closer both to the technology as well as the customer. This means they can better link technological possibilities to market needs. In addition to making better decisions, these decisions are made more rapidly because there are fewer levels of hierarchy. Beyond these "vertical advantages" that stem from few levels of hierarchy, are horizontal communication advantages. Allen (1977) found that the likelihood

engineers consult other employees is a function of physical proximity (which is closer in small companies). A more recent study finds this is still true even though researchers now have email and the internet (Liu 2016).

Given compelling arguments for either size to be more effective in conducting R&D, the issue of firm size and innovation has become an empirical matter. Cohen (2010) provides a comprehensive summary of that literature. The main stylized facts emerging from that summary are: 1) the likelihood of conducting R&D increases with firm size; 2) R&D rises monotonically with size (Baldwin and Scott 1987, Scherer and Ross 1990), and 3) R&D productivity (innovation per unit of R&D) tends to decline with firm size (Scherer 1965, Pavitt et al 1987, Acs and Audretsch 1988, 1990).

Thus the empirical literature has been unable to resolve the debate. Indeed the stylized facts introduce a puzzle of seemingly irrational behavior of large firms: R&D investment increases in scale, despite the fact that R&D productivity appears to decrease with scale. Accordingly, a new generation of theories has emerged to explain the empirical puzzle of large firm irrationality. These more nuanced theories don't advocate for a particular firm size. Rather they suggest the two sizes differ in the type of R&D they conduct, and that the returns to type of R&D vary with firm size. In essence they argue that choice of R&D strategy is endogenous--firms choose the strategy most likely to be effective given their size.

The first of the nuanced theories is that large firms are more likely to conduct basic research. Nelson (1959) argues this is due to a broader technological base, as well as a wider range of products firms are willing to produce if the research identifies opportunities. The second of the nuanced theories addresses the observation that radical innovation tends to decrease with firm size (Mansfield 1981). Rosen (1991) develops a model in which firms separately choose project riskiness and project scale. Because the scale of their output is greater, large firms prefer safer innovation that enhances their existing product lines. In contrast, precisely because they lack scale, small firms require riskier projects that have the potential for greater price premia. The final nuanced theory addresses the observation that process R&D tends to increase with firm size (Link 1982, Scherer 1991). Cohen and Klepper (1996) argue that because process innovation provides lower cost good/services to existing customers, the returns to process innovation increase in ex ante output. In contrast, because the returns to product innovation stem from licensing or sales to new customers, they are independent of ex ante

output. Accordingly, large firms favor process innovation, while small firms favor product innovation. Cohen and Klepper (1996) test their model using FTC Line of Business data, and confirm that process share of R&D increases with firm size. However, aside from the empirics in Cohen and Klepper (and the empirical observations motivating each theory), these more nuanced theories have not been tested.

Thus one explanation for the empirical puzzle of irrational firms, is that firm size determines the type of R&D, and that innovative outcomes rely on type. An alternative explanation for the empirical puzzle is that it is an artifact of weak measures of innovative outcomes. Typically, empirical studies rely on product counts or patent counts as the measure of R&D outcomes. However, if the type of innovation is endogenously determined by firm size as the more nuanced theories suggest, these tests may undercount large firm innovation. This is because process and incremental innovation are less likely to be patented. Indeed, Cohen and Klepper (1996) state they don't examine outcomes in their study of product versus product R&D, because process R&D is less likely to be patented.

Our goal in the paper is to resolve the empirical puzzle of irrational large firms. We hope to reconcile the empirical observation that R&D spending increases with scale, with the countervailing observation that R&D productivity decreases with scale. We employ two tacks—utilizing an alternative measure of R&D output, and testing the more nuanced theories of R&D. This requires both a more comprehensive measure of R&D outcome (to captures all forms of innovation), as well as data on R&D practices (to capture the innovation strategies in the more nuanced theories).

We exploit a recent measure of R&D productivity, RQ (short for Research Quotient) (Knott 2008) to pursue the first tack, and recently available data from the National Science Foundation (NSF) Business R&D and Innovation Survey (BRDIS) to pursue the second tack, in attempting to resolve the firm size puzzle. We expect to find either that the more comprehensive outcome measure indicates R&D productivity increases with scale, or that the returns to at least one form of innovation favored by small firms decreases with scale.

With regard to the first tack of employing a more comprehensive measure of innovative outcomes, we find that R&D productivity (measured as RQ) increases with scale. Thus the measurement tack offers one resolution of the firm size puzzle: R&D spending and RQ both increase with scale. We had limited success with the second tack of exploring the more nuanced

theories of firm size and innovation. While our results confirm that firm R&D behavior matches each theory: basic research, process innovation and incremental innovation all increase with scale, we could find no strategy favoring small firms. In particular, there were no strategies whose returns decreased in scale—there were merely strategies for which the small firm penalty was less severe. Taken together, the results indicate that large firms are rational. Moreover they suggest the prior puzzle of irrational firms stems from measurement problems, rather than from endogeneity issues.

Beyond offering a resolution to the puzzle of irrational large firms, which is important for theory, our results offer a broader implication for practice: Schumpeter (1942) appears to be correct that large firms are the chief engine of innovation (and accordingly economic growth). Not only do large firms (using the US Small Business Association definition of greater than 500 employees) conduct 5.75 more R&D in aggregate than small firms, they have 13% higher productivity with that R&D. Thus anti-trust policy may want to consider tradeoffs between the anti-competitive effects of large scale against the innovative benefits of large scale. Similarly, for firms, the prescription for them to behave more entrepreneurially (like small firms), should be treated with caution. At least one small firm strategy has lower returns to scale than the large firm counterpart.

The paper proceeds as follows. We first outline our empirical approach, providing greater details on the measurement tack (including the RQ measure itself), the nuanced theory tack, and the data. We next present results, then provide a summary and discuss implications.

II. Empirical Approach

This study is an effort to resolve the firm size puzzle by pursuing two tacks: 1) utilizing a new R&D output measure to see if the puzzle is merely an artifact of measurement problems, and 2) testing the more nuanced theories of firm size and R&D. Under the measurement tack, we utilize COMPUSTAT data on all publicly traded firms engaged in R&D. We first attempt to replicate the stylized fact that R&D productivity decreases with scale. We then repeat the test using the new RQ measure that captures the returns to R&D irrespective of form. Under the second tack of testing the more nuanced theories of firm R&D behavior, we exploit recently collected data on firms' R&D practices from the NSF's BRDIS survey. We first test

the impact of scale on the likelihood of employing each strategy. We then examine the impact of each strategy on R&D productivity.

2. 1 Replicating stylized facts

Our first set of empirics attempts to replicate the stylized facts that R&D increases with scale, while R&D productivity (measured as patents per unit of R&D) decreases with scale. Equation 1 examines the main effects of firm scale on behavior (R&D spending), while equation 2 models the main effects of scale on patent intensity. Both models include firm fixed effects and year effects. Our primary dataset for testing equation 1 comes from the COMPUSTAT North American Annual database. However to test equation 2, we need to merge that with data from the NBER patent database maintained by Hall, Jaffe and Trajtenberg (2001). The database provides detailed information about patent application and grant date, patent assignee, citation information for each patent and a link between patent assignee and COMPUSTAT GVKEY. This database starts in 1976 and ends at 2006.

Because we later utilize BRDIS data to examine firm behavior, we also test equations 1 and 2 using that data.

(1) $$\ln R_{it} = \beta_0 + \beta_1 \ln Y_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

(2) $$\textit{Patent Intensity}_{it} = \beta_0 + \beta_1 \ln Y_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

2.2 The Measurement tack

Once we replicate the stylized fact with COMPUSTAT data, we then retest equation 2, replacing patent intensity with a more comprehensive measure of R&D outcomes, RQ (short for research quotient), as shown in Equation 3.

(3) $$RQ_{it} = \beta_0 + \beta_1 \ln Y_{it} + \gamma_i + \delta_t + \varepsilon_{it}$$

2.2.1 The RQ measure

The RQ measure that we employ for the measurement tack, and another measure called TFP (total factor productivity) were both developed to circumvent problems with patent

measures for capturing firm innovation. The first concern with patent-based measures is they are not *universal*. In our sample, only 37% of the firms in COMPUSTAT who conduct R&D have any patents. Thus over 60% of firms conducting R&D will appear to have no innovation. The second concern with patents is they lack *uniformity*. Scherer and Harhoff (2000) report that 10% of U.S. patents account for 81%-85% of the economic value of all U.S. patents. To compensate for this, many studies weight patents by the number of citations they receive. However, Abrams et al (2013) show that even this correction may be problematic. Thus while citations may help mitigate the non-uniformity problem of patent value, they don't solve it.

TFP and RQ are similar in that they capture the source of technological change in the firm's production function. However, they differ in their assumptions about the source of that technological change. TFP maps onto Solow's (1957) theory of exogenous technological change—thus implicitly assumes firm innovative output is independent of its R&D. Accordingly, it is modeled as a shifter in the firm's production function, and empirically captured as a fixed effect or the residual in the firm's production function.

In contrast RQ, defined as the firm-specific output elasticity of R&D, maps onto models of endogenous technological change that treat R&D as investments made by profit-maximizing agents (Romer 1990, Thompson 1996, Lentz and Mortensen 2008). Accordingly, it is imbedded in the firm's production function as a continually evolving stock of knowledge, and is captured empirically as the output elasticity of R&D.

More macro-level versions of both measures have been used extensively in studies at the industry and economic level (See Hulten 2000 for a review of TFP studies; see Hall, Mairesse and Mohnen 2010 for a review of returns to R&D). The extension of these measures to the firm-level stems from the recognition there is tremendous heterogeneity among firms even within the same industry

Both TFP and RQ circumvent the universality and uniformity problems of patent-based measures. With respect to universality, both TFP and RQ are estimated from firms' financial data, so can be estimated for any firm doing R&D. With respect to uniformity, both measures are unitless (TFP is a shifter, RQ is an elasticity), so their interpretation is uniform across firms. However of the two measures, RQ is the more appropriate for tests of firm behavior and innovation because its foundational assumption (firms choose R&D to maximize firm value) matches the empirical reality that firms' invest in R&D to achieve growth. Beyond that, and

likely because of that, the most compelling advantage of RQ is *reliability*. Tests of RQ versus patent intensity and TFP indicate RQ is the only measure empirically consistent with propositions from models of endogenous growth theory (e.g., Thompson 1996, Lentz and Mortensen 2011) that R&D investment, market value and growth all increase in R&D productivity (Knott and Vieregger 2016). In contrast, patent intensity fails to hold under all three propositions, while total factor productivity (TFP) holds for the R&D investment and market value propositions, but not for the growth proposition.

Of course, RQ has limitations. In particular, it can only capture innovation derived from R&D. Thus it excludes innovation by the approximately 7% of non-R&D firms who report introducing product or service innovations in the prior three years (Boroush 2010). Further RQ can only measure aggregate innovation (the level at which you can form a production function). Thus it can't be used to measure productivity of particular projects. Finally, the fact that it is unitless means it can't be used to examine the knowledge content or flows of innovation (things patents are particularly well-suited to).

2.3 The Nuanced Theory Tack

To implement the nuanced theory tack, we first examine whether firm scale drives strategy in the manners predicted by theory. We then examine how the strategies affect R&D productivity. We utilize BRDIS data for both sets of examinations. These data have variables that map onto each of the nuanced theories, and also have financial variables that allow us to estimate RQ. The BRDIS variables mapping onto firm strategies in the nuanced theories are: portfolio horizon (basic, applied, development), which maps onto Nelson 1959, riskiness (new to market, new to firm, incremental), which maps onto Rosen 1991, and locus (product, service, process), which maps onto Cohen and Klepper 1996. For each strategy, we examine both the impact of scale on choice of that strategy (nine separate iterations of equation 4) and the impact of that strategy on outcome (nine separate iterations of equation 5).

(4) $$P(strategy)_i = \beta_0 + \beta_1 \ln Y_{ii} + \varepsilon_i$$

(5) $$RQ_{ij} = \beta_0 + \beta_{1i} strategy_j + \beta_2 \ln Y_{it} + \beta_{3j} strategy_j * \ln Y_i + \varepsilon_i$$

We first estimate equations 4 and 5 for each strategy by defining strategy as the nine uni-dimensional choices from the BRDIS questions (three each for horizon, riskiness and locus).

However, because it is unlikely firms choose the level of each uni-dimensional strategy in isolation, we also examine whether firms' R&D strategies are better characterized as bundles. Accordingly we use cluster analysis to construct strategy "archetypes" for portfolio horizon, riskiness, and locus.

We then repeat estimation of equations 4 and 5 replacing the uni-dimensional strategies with the strategy archetypes. Note that we have no expectations for these archetypes because there isn't theory to guide them. We merely use them as a robustness check of whether strategy affects the observed relationships between firm size and innovation.

2.4 Data and Variables

The data to support these tests come from four sources. The first tack utilizes data on publicly traded firms taken from a) the COMPUSTAT North American Annual database (1976 to 2012), the Wharton Research Data Services (WRDS) RQ database, and data on patent grants taken from the NBER patent database (1976 to 2006) maintained by Hall, Jaffe and Trajtenberg (2001). Data to support the second tack comes from the NSF's confidential BRDIS survey (2008-2011).

2.4.1 BRDIS Data

BRDIS is an annual survey of firms' R&D behavior conducted by the National Science Foundation (NSF) in conjunction with the U.S Census Bureau. BRDIS is a more expansive successor to the Survey of Industrial Research and Development (SIRD), which was conducted from fiscal years 1953 to 2007. Both surveys address the industry component of the NSF mandate "...to provide a central clearinghouse for the collection, interpretation, and analysis of data on scientific and engineering resources and to provide a source of information for policy formulation by other agencies of the Federal government."

The more expansive survey was deemed necessary because the bulk of R&D is now funded by industry, whereas in the 1950s, when the SIRD was created, the majority of R&D was funded by the US government. Thus greater insight into firm R&D behavior was warranted. In addition, the new survey better matches the Community Innovation Survey

(CIS) conducted by the EU countries.[2] Accordingly, BRDIS data support comparisons of innovative behavior between the US and other countries.

BRDIS is mailed annually to approximately 40,000 companies. The BRDIS sample is intended to represent the approximately 1.5 million for-profit companies in the United States with five or more domestic employees, both publicly or privately held. The overall response rate in the 2008 survey was 77.4%, and the response rate for the top 500 domestic R&D-performing companies was 92.6%. Of these responding firms, approximately 3% reported performing and/or funding R&D.

BRDIS gathers data on a number of R&D variables. The full surveys for each year are available at http://www.nsf.gov/statistics/srvyindustry/#qs, however we focus attention on the variables related to firm size *(ln(worldwide sales))*, the additional variables required to estimate RQ *(ln(worldwide R&D), ln(employees))* and firm R&D practices associated with the nuanced theories.

2.4.1.1 BRDIS R&D strategies data

BRDIS contains a number of questions related to firms' R&D strategies. We restrict attention to the three questions that best match the nuanced theories of firm size and R&D.

Theory 1: Large firms are more likely to conduct basic research

The BRDIS question that best addresses this theory, asks: "What percentage of R&D paid for and performed by your company was for:

a. *Basic research*: planned, systematic pursuit of new knowledge without specific immediate commercial application

b. *Applied research*: planned, systematic pursuit of new knowledge aimed at solving a specific problem or meeting a specific commercial objective

c. *Development*: the systematic use of research and practical experience to produce new or significantly improved goods, services or processes".

If Nelson (1959) is correct, we expect the answer to (a) to increase with firm size.

[2] The Community Innovation Survey (CIS) is a survey of innovation activity in enterprises in EU member states and ESS member countries. It is designed to provide information on the innovativeness of sectors by type of enterprises, on the different types of innovation, and on various aspects of the development of an innovation. http://ec.europa.eu/eurostat/web/microdata/community-innovation-survey

Theory 2: Radical innovation decreases with scale.

The BRDIS question that best addresses this theory, asks: "What percentage of this year's sales were from goods/services introduced in the past three years that were:

a. New (or significantly improved) to your market
b. New (or significantly improved) only to your company
c. Unchanged or only marginally modified"

The most radical innovations are goods/services that are new to the market (a). These have inherently higher uncertainty of commercial success than ones which are only new to the company (b). These in turn have higher uncertainty than incremental innovation (goods or services which are unchanged or only marginally modified). If Rosen is correct, we expect the answer to (a) to decrease with scale, while we expect the answer to (c) to increase with scale

Theory 3: Process innovation increases with scale; while product innovation decreases with scale.

The BRDIS question that best addresses this theory, asks: "Did your company introduce any of the following in the prior three years:

a. New (or significantly improved) goods
b. New (or significantly improved) services
c. New (or significantly improved) processes (methods of producing goods or services, distributions systems for inputs or outputs of your goods or services, support activities)"

If we interpret both new goods and new services as product innovation, and if Cohen and Klepper are correct, then the answers to (a) and (b) should decrease with scale, while the answer to (c) should increase with scale.

2.4.1.2 RQ estimation with BRDIS data

The RQ data to support test of equation 3 come from the WRDS RQ database. These RQ estimates are derived from the North American COMPUSTAT database. Thus we can match them to firm scale data from COMPUSTAT. In order to test equation 3 for firms in the BRDIS sample, we need to derive RQ from firms' financial data in BRDIS. We do so using

random coefficients estimation of the firm's production function in accordance with the general methodology described in Knott (2008). One complication with estimation using the BRDIS data is that its input variables differ from those collected from firms'10-K filings (the source for COMPUSTAT data on which the RQs in the WRDS RQ database were derived). While revenues, employees and R&D expenditures are the same in both datasets, BRDIS does not collect either capital or advertising. Accordingly we employ a model of the following form:

$$(6) \qquad \ln Y_{it} = (\beta_1 + \beta_{1i}) + (\beta_2 + \beta_{2i}) \ln L_{it} + (\beta_3 + \beta_{3i}) \ln R_{it-1} + (\beta_4 + \beta_{4i}) \ln S_{it-1} + \varepsilon_{it}$$

Firm level data items include (in $MM unless otherwise stated): *worldwide sales (Y_{it}),* *labor* as full-time equivalent employees (1000)*(L_{it}),* and *worldwide R&D (R_{it}).* From these primary data, we derive a secondary measure: firm-specific *spillovers (S_{it})* which follows the "density above" functional form of spillovers in Knott, Posen and Wu (KPW) (2009). The density-above form of spillovers sums the differences in knowledge (measured as R&D) between the focal firm and any firm with more knowledge than the focal firm. KPW found that this form of spillovers better predicted firm output that either a pooled form (all firms in an industry sharing all knowledge equally), or a leader-distance form (available spillovers equals the difference in knowledge between the focal firm and the knowledge leader).

We validated the abbreviated RQ estimation with COMPUSTAT data by comparing RQs estimated with the full set of inputs in Knott (2008) to those estimated with the restricted set of BRDIS inputs. That comparison indicates the RQs are 91.7% correlated across the full RQ equation and the limited equation (6).

A second restriction of the BRDIS data is that is has only been released for years 2008-2011. Accordingly the maximum window size per RQ estimate is four-years, whereas prior estimates of RQ (Knott 2008, Knott and Vieregger 2016) used seven and ten-year windows, respectively. Accordingly, we tested the impact of window size using COMPUSTAT data. Comparisons of RQs generated with COMPUSTAT data in a window surrounding the year 2000 indicate RQ estimates with a four-year window are 75.8% correlated with those estimated with seven-year window. Thus the shorter window will make it more difficult for us to obtain significant results. However if we do obtain significant results, the high correlation makes it likely results will hold and be more significant with the longer window.

Because RQ estimation consumes 4 years' data, we have only one observation for RQ (the 4 year mean). Accordingly, we form four-year averages of our scale variable (*worldwide sales*) and our outcome variables (*R&D* and *patent intensity*) as well as the strategy variables. This yields one observation per firm. Accordingly our analyses with BRDIS data form a cross-section rather than a panel.

Variables

As mentioned previously, our empirics comprise two datasets: 1) a dataset of public firms that combines variables from COMPUSTAT, the WRDS RQ database, and the NBER patent database, and 2) the confidential BRDIS dataset that is only accessible within US Census Research Data Centers (RDCs).

Our public dataset used to test equations 1-3 comprises three variables: firm scale, measured as *ln(Revenues)* from COMPUSTAT[3], *firm RQ* from the WRDS RQ database, and *patent intensity*. *Patent intensity* is a derived variable formed by dividing *patents granted* from the NBER patent database, by R&D from COMPUSTAT. This matches the "patents per unit of R&D" outcome measure in the stylized facts regarding firm size and innovation. A summary of the resulting dataset of public firms is provided in Table 1a.

We extract a number of variables from the BRDIS dataset. These include the variables necessary to compute RQ: *worldwide sales (Y_{it}), labor* as full-time equivalent employees (1000)*(L_{it}),* and *worldwide R&D (R_{it}),* the number of *patents*, from which we derive *patent intensity (patents/worldwide R&D).* In addition, we include answers to the questions pertaining to the three nuanced theories of firm R&D behavior: *Basic R&D (% of total R&D), Applied R&D (%),* and *Development (%), New to Market innovation (% of sales from), New to firm innovation (% of sales from),* and *Incremental innovation (% of sales from),* and dummy variables for the question: Did your firm introduce any of the following in the past three years (1=yes, 0=no): *Process R&D, Product R&D,* and *Service R&D.*

A summary of the resulting BRDIS dataset is provided in Table 1b.

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[3] While there are alternative measures of firm scale, such as assets and the number of employees, the nuanced theories of firm R&D behavior pertain to the scale of output rather than inputs.

Insert Table 1 about here

III. Results

3. 1 Replicating stylized facts

Our first set of empirics attempts to replicate the stylized facts that R&D increases with scale, while R&D productivity (measured as patents per unit of R&D) decreases with scale. Table 2 presents results for equation 1 which tests the impact of scale on behavior (R&D spending) (Model 1) and equation 2, which tests the impact of scale on patent intensity (Model 2) utilizing the dataset of public firms.

Regarding the impact of scale on R&D investment, results indicate that scale is positive and significant in explaining R&D investment. This matches stylized facts. Indeed, scale explains approximately 46% of intra-firm variance in R&D. The coefficient estimate of 0.529 implies that a 10% increase in the revenues increases R&D 5.3%. This is net of firm fixed effects and year effect.

Insert Table 2 about here

Regarding the impact of scale on outcomes, results in Model 2 indicate patent intensity is negative and significant. A 10% increase in revenue decreases patent intensity by 1.1 percentage points. This also matches stylized facts.

To check the robustness of these results, we subdivided the sample into large firms and small firms, using the SBA threshold of 500 employees. Models 3 and 4 present results for small firms, while models 5 and 6 present results for large firms. In both cases, R&D investment increases with scale, while R&D productivity (measured as patent intensity) decreases with scale.

Because we later utilize BRDIS data to examine firm behavior, we also test equations 1 and 2 using that data. These results (presented in models 7 and 8) match those for the public sample. Thus across all samples and subsamples, we are able to replicate the puzzle that R&D investment increases with scale, while R&D productivity decreases with scale.

3.2 The Measurement tack

Our first effort to resolve the puzzle of irrational large firms, employs a measurement tack to see if the puzzle is merely an artifact of measurement problems. To examine that, we retest equation 2, replacing patent intensity with the more comprehensive measure of R&D outcomes, RQ (equation 3).

The results for the measurement tack are presented in Table 3. Model 1 utilizes the full set of public firms, while Models 2-4 utilize the small firm subsample, the large firm subsample, and the BRDIS sample, respectively. All models indicate that R&D productivity (measured as RQ) increases significantly with scale. The coefficient on scale is fairly consistent across the models (0.02), meaning that a 10% increase in revenues increases RQ by 0.02 points (roughly 20% of the mean value for RQ in the public sample.

Insert Table 3 about here

Thus our results using the RQ measure yield firm behavior and outcomes that are consistent with one another: R&D investment and R&D productivity both increase with scale. While these results resolve the prior puzzle of inconsistency between behavior and returns, the results may be biased by the fact that R&D strategy varies with size, and that returns to R&D strategies may also vary with size. Accordingly, we begin to decompose that by looking separately at the impact of scale on choice of strategy and the impact of strategy on behavior and outcomes (the second tack).

3.3 The Nuanced Theory Tack

To explore the second tack, we next examine firms' R&D strategies. As mentioned previously, these strategies map onto the three nuanced theories of how firm scale affects behavior and outcomes. These strategy data are only available in BRDIS, so we utilize that dataset for all these tests. We examine both the impact of scale on choice of strategy (equation 4) and the impact of strategy on outcomes (equation 5).

3.3.1. Choice of R&D Strategy

Results for tests of the impact of scale on choice of R&D strategy (Table 4) provide support

for all three theories of scale-based R&D strategy. The percentage of R&D devoted to basic research is increasing in scale (consistent with Nelson 1959) (theory 1). The likelihood of introducing a process innovation is also increasing in scale (consistent with Cohen and Klepper 1996) (theory 2). Note however that the likelihood of introducing product and service innovations is also increasing in scale (though the likelihood of process innovation is 2.5 times that for product innovation). The result that all forms of innovation are increasing in scale can be explained by measure coarseness. The measure equals one if *any* innovation was introduced in the prior three years. Thus it can't be used to gauge the share of process R&D relative to product R&D.

Insert Table 4 about here

Finally, the extent of incremental innovation is increasing in scale, while the extent of radical innovation is decreasing in scale. Both results support Rosen's (1991) theory regarding the impact of scale on the riskiness of R&D (theory 3).

Thus all three theories correctly anticipate the how scale affects choice of R&D strategy. This is not a surprise since all three theories were motivated by empirical observations.

3.3.2 Impact of R&D Strategy on Outcomes.

Having demonstrated that firm R&D behavior is consistent with theories of strategic choice, we now turn to the question of returns (RQ) to those strategies (Table 4). Each model in Table 5 captures the impact of the strategy identified in the column header. For example, model 1 captures the impact of basic research on RQ. We allow each strategy to have both intercept and slope (with scale) effects. Table 5 reveals that across the set of strategies, the main effects of scale continue to be significant and of similar magnitude to the main model of scale in Table 2 (0.014). While the coefficients on strategy tend to be significant, the signs on the intercept and slope terms consistently have opposite sign. Thus the role of strategy is to slightly rotate the relationship between scale and RQ as shown in Figure 1 for the three horizon strategies. However for all strategies, the net returns to R&D increase with scale (with slopes ranging from 0.018 for service innovation to 0.022 for development). Accordingly, there is no strategy favoring small firms. Rather there are merely strategies for which the penalty for small scale is less severe.

Insert Table 5 about here

Insert Figure 1 about here

Thus while scale is significant in explaining firms' R&D strategies, those strategies appear to have little power in explaining the differential R&D productivity of large and small firms. The primary driver of R&D productivity continues to be scale after we take these strategies into account.

3.3.3 Strategy archetypes

The prior section formally tests the nuanced theories of how firm scale affects R&D behavior and outcomes. Each of these theories was uni-dimensional, e.g., scale increases x, or scale decreases y. However it seems likely that firms use combinations of these strategies, i.e., firms don't just choose the level of basic research, they jointly choose the proportions of basic research, applied research and development. While there is no theory to guide us here, we wanted to see if considering joint strategies improved the performance of firm strategy in explaining R&D productivity.

3.3.3.1 Defining Strategy archetypes.

To generate the strategy archetypes, we conducted cluster analysis of the three uni-dimensional strategies within each main strategy category (horizon, riskiness and form). We followed a two-step process. In the first step, we created nine separate groupings, ranging from a minimum of two clusters to a maximum of ten, and compare the Calińsky and Harabasz pseudo-F index across each grouping. Because the F-scores are similar across several groupings, in a second step we examined the content of each grouping in selecting our final cluster sizes. The objective of this second step was to ensure that the automatically-generated clusters also conform to observed strategic groupings in management practice. While our reported clusters are based on kmeans partition cluster analysis, we also tested groupings based on kmedians, agglomerative hierarchical clustering methods, and the examination of dendograms, which produced qualitatively similar

results. The analysis yielded five archetypes for horizon, six for riskiness and seven for locus. These archetypes are characterized in Table 6.

Insert Table 6 about here

Table 6a reveals that the horizon strategy clustered into five archetypes: BAD1 through BAD5.[4] Each archetype has a fairly intuitive interpretation. BAD1: principally development, with some applied research; BAD2: applied/development balance; BAD3: exclusively development; BAD4: principally basic research; and BAD5: principally applied research. Note that BAD4 and BAD5 are rare, comprising 30 and 60 firms, respectively, while BAD3 is most prevalent (50% of firms).

Table 6b reveals that the riskiness strategy clustered into six archetypes: OWN1 through OWN6.[5] Each archetype has a fairly intuitive interpretation. OWN1: principally new to market; OWN2: principally new to firm; OWN3: new to market/incremental balance; OWN4: New to firm/incremental balance; OWN5: exclusively incremental; OWN6: principally incremental. OWN1 and OWN2 are relatively rare, comprising 90 and 70 firms, respectively, while OWN5 is most prevalent with 47% of firms.

Finally, Table 6c reveals that the locus strategy clustered into seven archetypes: PSP1 through PSP6.[6] As with both the horizon riskiness strategies, each archetype has a fairly intuitive interpretation. PSP1: balance across product, service and process innovation; PSP2: exclusively product; PSP3: exclusively process; PSP4: exclusively service; PSP5: product/service balance; PSP6: product/process balance; PSP7: service/process balance. PSP3, 4, 6 and 7 are relatively rare, comprising 10, 20, 80 and 60 firms, respectively, while PSP1 is most prevalent with 52% of firms.

3.3.3.2 Impact of Scale on Choice of Strategy Archetypes.

We repeated the tests of the impact of scale on choice of strategy (equation 4) and the impact of strategy on outcomes (equation 5) replacing the uni-dimensional strategies in Tables 3 and 4 with the strategy archetypes.

Table 7 reveals that firm choice regarding strategy archetypes resembles that for uni-

[4] The abbreviation is derived from basic, applied, and development.
[5] The abbreviation stems from the question format asking about own-R&D investments.
[6] The abbreviation is derived from products (or goods), services, and processes.

dimensional strategies in Table 3. In particular, strategies dominated by process (PSP1) and incremental innovation (OWN5) increase with scale, while strategies with substantial radical (OWN1, OWN3) and product innovation (PSP2, PSP5, PSP6) decrease with scale.

Insert Table 7 about here

Impact of Strategy Archetypes on RQ Looking next at the impact of strategy on outcomes (Table 8), we again obtain results similar to those for uni-dimensional strategy in Table 5. In particular, there is no strategy whose returns decrease with scale. Thus whether we employ the uni-dimensional strategies from the nuanced theories or the archetypal strategies, these strategies play only a minor role in explaining R&D productivity. Scale remains the primary driver of R&D productivity. Further for either the uni-dimensional strategies or the archetypal strategies, there is no strategy whose returns decrease with scale. There are merely strategies for which the penalty for small scale is less severe.

Insert Table 8 about here

IV. Conclusion

There has been a long-standing debate regarding the impact of firm size on innovation. Stylized facts indicate R&D spending increases with size, while R&D productivity decreases with size. This creates a puzzle of large firm irrationality. Three theories have been advanced to resolve the puzzle. Each theory proposes that firm size affects the type of R&D firms conduct, and that the productivity of types differs with firm size. Until recently, we have been unable to comprehensively test these theories because we lacked firm-level data on the type of R&D firms conduct, and because the most common measures of R&D outcomes (product/patent counts) embed type. Accordingly, a second possible resolution of the puzzle is that it is merely an artifact of poor outcome measures, that should disappear with a more robust measure.

We exploited two recent developments to explore both potential resolutions. In particular,

we utilized a recently created dataset (BRDIS) to pursue the nuanced theory tack; and utilized a recent measure of R&D productivity (RQ) to pursue the measurement tack. Before pursuing either tack however, we first replicated the stylized facts that R&D spending increases with scale while R&D productivity (measured as patent intensity) decreases with scale. We did so using both a public dataset and the confidential BRDIS dataset.

We then tested whether results changed when utilizing RQ as the measure of R&D productivity, and found indeed that R&D productivity increased with scale. These results held not only with the full sample of public firms, but with large and small firm subsamples of the public dataset, as well as with the BRDIS dataset. Thus the measurement tack offers one resolution of the large firm irrationality puzzle—R&D investment and R&D productivity both increase with scale when we use a more robust measure of R&D productivity.

We next examined whether the nuanced theories of how scale affects R&D behavior offered an additional resolution of the puzzle of large firm irrationality. We found first that firm size does indeed affect choice of R&D strategy as expected in each of the theories proposed by Nelson (1959), Rosen (1991) and Cohen & Klepper (1996). All three theories correctly anticipated the strategies preferred by large firms. In particular, scale increased the likelihood of incremental R&D (consistent with Rosen, 1991), process R&D (consistent with Cohen and Klepper, 1996) and basic research (consistent with Nelson 1959).

Looking next at the impact of those strategies on R&D productivity, we found that none could resolve the puzzle. While all the theories correctly anticipated the relationships between firm size and R&D strategy, we found that none of the small firm strategies favored small firms—the productivity of all strategies increased with scale. Rather, there were merely strategies that penalized small scale less.

There are limitations to these results. In particular, while the results for the measurement tack are robust, in that they span all the samples and subsamples, and include fixed effects and year effects, the results for the nuanced theory tack all rely on cross-sectional data. This is not because BRDIS is inherently cross-sectional, it is because estimating RQ requires multiple years of observations, and this estimation consumed all years of BRDIS data that we had access to. It is remotely possible that the results regarding the impact of practices on R&D productivity may change with panel data. An additional limitation pertains to the product/process strategy results.

The BRDIS measures (yes/no) don't map cleanly onto theory (share of R&D). Accordingly, there may be stronger results for process versus product innovation if we had a better measure.

Beyond reconciling the puzzle of irrational large firms, which is important for theory, our results offer a broader implications for practice: Schumpeter (1942) appears to be correct that large firms are the chief engine of innovation (and accordingly economic growth). Not only do large firms (using the US Small Business Association definition of greater than 500 employees) conduct 5.75 more R&D in aggregate than small firms, they have 13% higher productivity with that R&D. However this merely captures the private returns to their R&D. A further benefit of large firm R&D is that it generates the spillovers upon which small firm innovation free-rides.

Thus anti-trust policy may want to consider tradeoffs between the anti-competitive effects of large scale against the innovative benefits of large scale. Similarly, for firms, the prescription for them to behave more entrepreneurially (like small firms), should be treated with caution. At least one small firm strategy (radical innovation) has lower returns to scale than the large firm counterpart (incremental innovation).

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TABLE 1. Data Summary

Table 1a. Data on Public Firms
Observations=32608

Variable	Mean	Std. Dev.
ln(Revenue)	5.59	2.55
ln(R&D)	2.32	2.41
RQ (7-year)	0.13	0.12
Patent intensity	0.02	0.05

Table 1b. BRDIS Data Summary
Observations=2030 (rounded)

Variable	Mean	Std. Dev.
Sales (log)	11.80	2.14
Employees (log)	6.19	1.90
R&D (log)	8.60	1.95
Spillover (log)	12.45	4.57
R&D Intensity	0.26	2.99
RQ	0.26	0.05
Basic R&D (percent)	4.20	9.07
Applied R&D (percent)	14.71	17.81
Development R&D (percent)	81.09	20.83
Process R&D (percent)	0.58	0.37
Product R&D (percent)	0.77	0.32
Service R&D (percent)	0.33	0.37
R&D Incremental (percent)	76.88	25.61
R&D New-to-Market (percent)	12.02	19.08
R&D New-to-Firm (percent)	11.10	17.38
Patents granted 2008	15.56	116.75

* All variables calcuated as 4-year average of BRDIS data

TABLE 2. Replicating Stylized Facts

	Public data						BRDIS data	
	All firms		Large (>500 employees)		Small (<500 employees)			
Dependent Variables:	ln(R&D)	Patent intensity	ln(R&D)	Patent intensity	ln(R&D)	Patent intensity	ln(R&D)	Patent intensity
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
ln(Sales)	0.529***	-0.011***	0.846***	-0.010***	0.244***	-0.024*	0.621***	-0.0005*
	(0.027)	(0.002)	(0.029)	(0.002)	(0.026)	(0.011)	(0.019)	(0.0002)
Constant	-1.520***	0.061***	-3.048***	0.056***	-1.111*	0.316**	1.263***	0.0078*
	(0.123)	(0.011)	(0.167)	(0.012)	(0.448)	(0.092)	(0.2194)	(0.0028)
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	No	No
R-squared	0.458	0.086	0.633	0.105	0.141	0.138	0.463	0.006
N	32608	13400	22051	12368	10557	1032	2030	900

TABLE 3. Test of Measurement Tack

Dependent Variable: RQ				
	Public data			BRDIS data
	All firms	Large	Small	
	Model 1	Model 2	Model 3	Model 4
ln(Sales)	0.017***	0.020***	0.019***	0.014***
	(0.004)	(0.004)	(0.006)	(0.001)
Constant	0.187***	0.166***	0.119	0.094***
	(0.024)	(0.026)	(0.095)	(0.006)
Fixed effects	Yes	Yes	Yes	No
Year effects	Yes	Yes	Yes	No
R-squared	0.090	0.192	0.031	0.412
N	32608	22051	10557	2030

TABLE 4. Impact of Scale on Choice of Uni-dimensional R&D Strategy

	Dependent Variable								
	% Basic	% Applied	% Develop-ment	Prob: Product	Prob: Service	Prob: Process	% Sales New-to-Market	% Sales New-to-Firm	% Sales Incre-mental
ln(Sales)	**0.760**	0.221	-0.338	**0.086**	**0.177**	**0.219**	**-1.047**	-0.329	**1.624**
SE	0.177	0.242	0.275	0.043	0.022	0.032	0.264	0.249	0.306
Constant	**-11.704**	**8.386**	**88.680**	**1.614**	**-1.802**	**-0.865**	**19.629**	**10.441**	**59.359**
SE	2.177	2.917	3.321	0.503	0.266	0.364	3.177	3.000	3.667
R2	0.002	0.000	0.000	0.004	0.024	0.029	0.001	0.000	0.002
Observations	2030	2030	2030	2030	2030	2030	2030	2030	2030

TABLE 5. Impact of Uni-dimensional R&D Strategy On RQ

Dependent Variable: RQ	Each Model Represents a Different Strategy (N=2030)								
	% Basic	% Applied	% Develop-ment	Prob: Product	Prob: Service	Prob: Process	% Sales New-to-Market	% Sales New-to-Firm	% Sales Incremen-tal
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
Strategy	-0.0006	**-0.0013**	**0.0010**	**0.0850**	**0.0610**	**0.0720**	**-0.0010**	**0.0006**	**0.0006**
	(0.0004)	(0.0002)	0.0002	(0.0140)	(0.0120)	(0.0130)	(0.0001)	(0.0003)	(0.0001)
Strategy*ln(Sales)	0.0000	**0.0001**	**-0.0001**	**-0.0070**	**-0.0070**	**-0.0070**	**0.0001**	**-0.0001**	**0.0000**
	(0.0001)	(0.0001)	0.0000	(0.0010)	(0.0009)	(0.0010)	(0.0001)	(0.0001)	(0.0001)
ln(Sales)	**0.0142**	**0.0126**	**0.0220**	**0.0200**	**0.0180**	**0.0190**	**0.0130**	**0.0150**	**0.0164**
	(0.0004)	(0.0005)	0.0010	(0.0001)	(0.0005)	(0.0008)	(0.0004)	(0.0004)	(0.0010)
Constant	**0.0972**	**0.1160**	0.0080	**0.0280**	**0.0620**	**0.0470**	**0.1140**	**0.0880**	**0.0520**
	(0.0050)	(0.0060)	0.0160	(0.0110)	(0.0060)	(0.0090)	(0.0050)	(0.0050)	(0.0120)
R-squared	0.4120	0.4210	0.4200	0.4230	0.4420	0.4230	0.4340	0.4170	0.4300

TABLE 6. Strategy Archetypes

		CLUSTER MEANS (STDEV)		
	Firms	BASIC %	APPLIED %	DEVELOPMENT %
BAD1	630	5.13	16.96	77.91
		6.75	*6.26*	*6.32*
BAD2	290	8.12	37.42	54.46
		8.60	*9.65*	*8.64*
BAD3	1020	0.89	2.85	96.27
		2.31	*3.60*	*4.37*
BAD4	30	51.17	14.53	34.30
		23.91	*13.78*	*19.89*
BAD5	60	5.57	79.79	14.63
		8.27	*15.29*	*12.09*
TOTAL	2030	4.20	14.71	81.09
		9.07	*17.81*	*20.83*

	Firms	NEW TO MARKET %	NEW TO FIRM %	INCREMENTAL %
OWN1	90	81.01	6.08	12.91
		13.93	*9.80*	*11.15*
OWN2	70	12.31	78.94	8.75
		17.19	*20.16*	*11.10*
OWN3	160	39.77	14.29	45.94
		9.90	*12.96*	*12.53*
OWN4	180	8.70	33.54	57.76
		8.04	*10.14*	*10.35*
OWN5	960	2.17	2.48	95.35
		2.92	*3.19*	*4.10*
OWN6	570	11.25	9.82	78.93
		7.94	*6.85*	*5.76*
TOTAL	2030	12.02	11.10	76.88
		19.08	*17.38*	*25.61*

	Firms	PRODUCT	SERVICE	PROCESS
PSP1	1050	0.81	0.70	0.56
		0.29	*0.32*	*0.32*
PSP2	190	0.73	0.00	0.00
		0.30	*0.00*	*0.00*
PSP3	10	0.00	0.00	0.50
		0.00	*0.00*	*0.30*
PSP4	20	0.00	0.47	0.00
		0.00	*0.26*	*0.00*
PSP5	620	0.84	0.64	0.00
		0.24	*0.30*	*0.00*
PSP6	80	0.69	0.00	0.49
		0.31	*0.00*	*0.28*
PSP7	60	0.00	0.67	0.68
		0.00	*0.28*	*0.32*
TOTAL	2030	0.77	0.33	0.58
		0.32	0.37	0.37

TABLE 7. Impact of Scale on Choice of R&D Strategy Archetypes

Dependent Variable: Different Strategies (N=2030)

	BAD1	BAD2	BAD3	BAD4	BAD5
ln(Sales)	0.041	0.018	-0.026	-0.067	-0.130
SE	0.022	0.031	0.021	0.104	0.079
Constant	**-1.282**	**-2.000**	**0.301**	**-3.325**	**-1.960**
SE	0.271	0.369	0.250	1.208	0.901
Prob>chi2	0.070	0.563	0.220	0.521	0.101

	OWN1	OWN2	OWN3	OWN4	OWN5	OWN6
ln(Sales)	**-0.231**	-0.123	**-0.101**	-0.006	**0.081**	-0.001
SE	0.091	0.068	0.042	0.038	0.021	0.022
Constant	**-0.488**	**-1.890**	**-1.272**	**-2.247**	**-1.068**	**-0.918**
SE	1.001	0.782	0.488	0.452	0.250	0.269
Prob>chi2	0.011	0.071	0.017	0.865	0.000	0.958

	PSP1	PSP2	PSP3	PSP4	PSP5	PSP6	PSP7
ln(Sales)	**0.201**	**-0.220**	-0.081	-0.092	**-0.105**	**-0.207**	-0.046
SE	0.022	0.036	0.091	0.084	0.022	0.049	0.078
Constant	**-2.295**	**0.211**	**-4.266**	**-3.354**	**0.408**	**-0.787**	**-3.019**
SE	0.263	0.407	1.087	0.985	0.263	0.559	0.920
Prob>chi2	0.000	0.000	0.372	0.274	0.000	0.000	0.555

TABLE 8. Impact of R&D Strategy Archetypes on RQ

Dependent variable: RQ (N=2030)

	BAD1	BAD2	BAD3	BAD4	BAD5
Strategy	-0.001	*0.005*	0.000	**-0.012**	-0.003
	0.002	0.002	0.002	0.006	0.007
ln(Sales)	**0.014**	**0.014**	**0.014**	**0.014**	**0.014**
	0.001	0.000	0.001	0.001	0.001
Constant	**0.095**	**0.094**	**0.094**	**0.095**	**0.094**
	0.006	0.006	0.006	0.006	0.006
R-squared	0.412	0.413	0.412	0.413	0.412

	OWN1	OWN2	OWN3	OWN4	OWN5	OWN6
Strategy	**-0.026**	-0.008	-0.002	**-0.006**	**0.010**	-0.002
	0.005	0.005	0.003	0.003	0.002	0.002
ln(Sales)	**0.014**	**0.014**	**0.014**	**0.014**	**0.014**	**0.014**
	0.001	0.001	0.001	0.001	0.001	0.001
Constant	**0.098**	**0.095**	**0.096**	**0.095**	**0.092**	**0.095**
	0.006	0.006	0.006	0.006	0.006	0.006
R-squared	0.424	0.413	0.412	0.413	0.422	0.412

	PSP1	PSP2	PSP3	PSP4	PSP5	PSP6	PSP7
Strategy	**-0.007**	0.004	-0.007	-0.004	**0.010**	-0.005	**-0.013**
	0.001	0.003	0.008	0.008	0.002	0.004	0.004
ln(Sales)	**0.015**	**0.014**	**0.014**	**0.014**	**0.015**	**0.014**	**0.014**
	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Constant	**0.094**	**0.093**	**0.094**	**0.094**	**0.089**	**0.095**	**0.094**
	0.006	0.006	0.006	0.006	0.006	0.006	0.006
R-squared	0.417	0.412	0.412	0.412	0.421	0.412	0.414

FIGURE 1. Net Effect of Scale on RQ for Horizon Strategies

